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DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Parametric surfaces II: Tangent Plane

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0.1 Introduction

Let $\mathcal{S}$ be a regular surface described by a vector function $\vec{r}: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$. Let $(a, b, c) \in \mathcal{S}$ and let $(s_0, t_0) \in D$ (actually from the definition of regular surface, we take (a, b, c) so that (s_0, t_0) lies in the interior of the domain D) be such that $(a, b, c) = \vec{r}(s_0, t_0)$. The scope in this section is to write down the cartesian equation of the tangent plane to the regular surface $\mathcal{S}$ at the point (a, b, c) .

Definition 0.1 (Tangent vector to a surface)

We say that a vector $\vec{u} \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$ is tangent to the surface $\mathcal{S}$ at the point $(a, b, c) \in \mathcal{S}$ if there exists a regular curve $\mathcal{C}$ described by a vector function $\vec{\gamma}: (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^3$ such that:

- (i) $\vec{\gamma}(\tau) \in \mathcal{S} \quad \forall \tau \in (-\varepsilon, \varepsilon)$;
- (ii) $\vec{\gamma}(\tau_0) = (a, b, c)$ for some $\tau_0 \in (-\varepsilon, \varepsilon)$;
- (iii) $\vec{\gamma}'(\tau_0) = \vec{u}$.

The set of all tangent vector to the surface $\mathcal{S}$ at the point (a, b, c) is called the **tangent plane to the surface $\mathcal{S}$ at the point (a, b, c)** .

Now to the aim of finding the cartesian equation of the tangent plane to the parametric curve $\mathcal{S}$ at the point $(a, b, c) = \vec{r}(s_0, t_0)$, we fix $t = t_0$ and consider the curve $\mathcal{C}_1$ described by the vector function $\vec{u}(s) := \vec{r}(s, t_0)$ which is continuously differentiable and also, from the regularity of the surface $\mathcal{S}$ it follows that $\vec{u}'(s) = \vec{r}_s(s, t_0) \neq (0, 0, 0)$ for every s close to s_0 . Therefore, the curve $\mathcal{C}_1$ is regular around the point (a, b, c) with the direction of its tangent line at that point vector being $\vec{u}'(s_0) = \vec{r}_s(s_0, t_0)$. Therefore, the vector $\vec{r}_s(s_0, t_0)$ is tangent to the surface $\mathcal{S}$ according to the definition 0.1.

Similary, fixing $s = s_0$ we will obtain that the vector $\vec{r}_t(s_0, t_0)$ is also tangent to the surface $\mathcal{S}$.

Now, from the regularity of the surface $\mathcal{S}$ we have that

$$\vec{r}_s(s_0, t_0) \times \vec{r}_t(s_0, t_0) \neq (0, 0, 0).$$

Therefore, the two vectors $\vec{r}_s(s_0, t_0)$ and $\vec{r}_t(s_0, t_0)$ determine the tangent plane to the surface $\mathcal{S}$ at the point (a, b, c) and its normal vector is

$$\vec{r}_s(s_0, t_0) \times \vec{r}_t(s_0, t_0) \neq (0, 0, 0). \quad (1)$$

Now, having the normal vector to the tangent plane and having the point (a, b, c) which lies on that plane, we are in a position to write down the cartesian equation of the plane.

Example 0.2 Let $\mathcal{S}$ be the surface of the cone described by the vector function $\vec{r}: [0, 1] \times [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(s, t) := (s \cos t, s \sin t, hs) \quad \forall (s, t) \in [0, 1] \times [0, 2\pi].$$

Write down the cartesian equation of the tangent plane to the surface $\mathcal{S}$ at the point $\left(-\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{h}{2}\right)$.

Solution: One can easily find that the point $\left(-\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{h}{2}\right)$ corresponds to $s = \frac{1}{2}$ and $t = \frac{5\pi}{6}$. The normal vector to the tangent plane is given by

$$\vec{r}_s(1/2, 5\pi/6) \times \vec{r}_t(1/2, 5\pi/6).$$

So we need to find

$$\vec{r}_s(s, t) = (\cos t, \sin t, h) \quad \text{and} \quad \vec{r}_t(s, t) = (-s \sin t, s \cos t, 0).$$

So,

$$\vec{r}_s(1/2, 5\pi/6) \times \vec{r}_t(1/2, 5\pi/6) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\sqrt{3}/2 & 1/2 & h \\ -1/4 & -\sqrt{3}/4 & 0 \end{vmatrix} = (h\sqrt{3}/4, -h/4, 1/2).$$

Therefore, the tangent plane to the surface at point $\left(-\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{h}{2}\right)$ has cartesian equation

$$\begin{aligned} \frac{h\sqrt{3}}{4}\left(x + \frac{\sqrt{3}}{4}\right) - \frac{h}{4}\left(y - \frac{1}{4}\right) + \frac{1}{2}\left(z - \frac{h}{2}\right) &= 0 \Leftrightarrow \frac{h\sqrt{3}}{2}\left(x + \frac{\sqrt{3}}{4}\right) - \frac{h}{2}\left(y - \frac{1}{4}\right) + \left(z - \frac{h}{2}\right) = 0 \\ &\Leftrightarrow \frac{h\sqrt{3}}{4}x - \frac{h}{2}y + z = 0. \end{aligned}$$

Exercises 0.3

- Consider the surface $\mathcal{S}$ described by the vector function $\vec{r}: [0, 1] \times [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(s, t) := (2 \cos t \sin s, \sin t \sin s, 3 \cos s) \quad \forall (s, t) \in [0, 3\pi/2] \times [0, \pi].$$

- Show that the surface $\mathcal{S}$ is regular.
- Write down the cartesian equation of the tangent plane to the surface $\mathcal{S}$ at the point $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{4}, \frac{3\sqrt{3}}{2}\right)$.

- Let $\mathcal{S}$ be the surface described by the vector function $\vec{r}: [\frac{1}{3}, 3] \times [0, \frac{3\pi}{2}] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(s, t) := (s^2 \cos t + 2, 2s + t, s^2 \sin t + 1) \quad \forall (s, t) \in [1/3, 3] \times [0, 3\pi/2].$$

- Show that the surface $\mathcal{S}$ is regular.
- Write down the cartesian equation of the tangent plane to the surface $\mathcal{S}$ at the point $(1, 2 + \pi, 1)$.

0.2 The particular case of a level surface

0.2.1 Level curves of differentiable functions

We recall that the level curves of a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ are the sets

$$\mathcal{C}_k := \{(x, y) \in \mathbb{R}^2 : g(x, y) = k\} \quad k \in \mathbb{R}.$$

Whenever $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a continuously differentiable function, we intend to find the cartesian equation of the tangent line to the level curve $\mathcal{C}_k$ at a point $(a, b) \in \mathcal{C}_k$. To this aim, we will assume that $\nabla g(a, b) \neq (0, 0)$ and hence, that the piece of the curve $\mathcal{C}_k$ around the point (a, b) is regular and is described by the vector function $\vec{r} : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^2$ with $\vec{r}(0) = (a, b)$. In that case, we recall that the vector $\vec{r}'(0)$ is the direction of the tangent line to the curve $\mathcal{C}_k$ at the point (a, b) .

We get that

$$g(\vec{r}(t)) = k \quad \text{for every } t \in (-\varepsilon, \varepsilon). \quad (2)$$

So, if we differentiate both sides of (2) w.r.t. t we get

$$\nabla g(\vec{r}(t)) \cdot \vec{r}'(t) = 0 \text{ for every } t \in (-\varepsilon, \varepsilon) \Rightarrow \nabla g(\vec{r}(0)) \cdot \vec{r}'(0) = 0 \Rightarrow \nabla g(a, b) \perp \vec{r}'(0).$$

Since the vector $\vec{r}'(0)$ is the direction of the tangent line to the curve $\mathcal{C}_k$ at the point (a, b) , it follows that $\nabla g(a, b)$ is perpendicular to that tangent line. We are then able to write down the cartesian equation of the tangent line to the level curve at the point (a, b) . In fact,

$$\underbrace{\text{The point } (x, y) \text{ belongs to the tangent line if and only if}}_{\text{Scalar product}} \quad \underbrace{(x - a, y - b) \cdot \nabla g(a, b) = 0}_{\text{The cartesian equation of the tangent line at } (a, b)} \Leftrightarrow \underbrace{g_x(a, b)(x - a) + g_y(a, b)(y - b) = 0}_{\text{The cartesian equation of the tangent line at } (a, b)}. \quad (3)$$

0.2.2 Level surfaces of differentiable functions

We recall that the level surfaces of a function $h : \mathbb{R}^3 \rightarrow \mathbb{R}$ are the sets

$$\mathcal{S}_k := \{(x, y, z) \in \mathbb{R}^3 : h(x, y, z) = k\} \quad k \in \mathbb{R}.$$

Whenever, $h : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a continuously differentiable function, we intend to find the cartesian equation of the tangent plane to the level surface $\mathcal{S}_k$ at a point $(a, b, c) \in \mathcal{S}_k$ with $\nabla h(a, b, c) \neq (0, 0, 0)$. It follows from (i) in Definition 0.1 that

$$\underbrace{h(\vec{\gamma}(\tau)) = k \quad \forall t \in (-\varepsilon, \varepsilon)}_{\text{Differentiating (CR)}} \Rightarrow \underbrace{\nabla h(\vec{\gamma}(\tau)) \cdot \vec{\gamma}'(\tau) = 0}_{\tau = \tau_0} \Rightarrow \nabla h(a, b, c) \cdot \vec{u} = 0. \quad (4)$$

Therefore, $\nabla h(a, b, c)$ is orthogonal to every tangent vector to the level surface $\mathcal{S}_k$ at the point (a, b, c) . So, $\nabla h(a, b, c)$ is orthogonal to the tangent plane to $\mathcal{S}_k$ at the point (a, b, c) and hence, is the normal vector to that plane. Thus, that plane has cartesian equation

$$\Leftrightarrow \underbrace{(x - a, y - b, z - c) \cdot \nabla h(a, b, c) = 0}_{\text{Scalar product}} \Leftrightarrow \underbrace{h_x(a, b, c)(x - a) + h_y(a, b, c)(y - b) + h_z(a, b, c)(z - c) = 0}_{\text{The cartesian equation of the tangent plane at } (a, b, c)}. \quad (5)$$

Example 0.4

Write down the cartesian equation to the surface $\mathcal{S}$ with equation $-3x^2yz - x^3 + 2y^3 + xz + z^2 = -9$ at the point $(1, -2, 1)$.

Solution: We consider the function $h(x, y, z) = -3x^2yz - x^3 + 2y^3 + xz + z^2$. So, the surface $\mathcal{S}$ is a level surface of the function f and therefore the normal vector of the tangent plane to that surface at the point $(1, -2, 1)$ is given by $\nabla h(1, -2, 1)$. We find that

$$\nabla h(x, y, z) = (-6xyz - 3x^2 + z, -3x^2z + 6y^2, -3x^2y + x + 2z) \Rightarrow \nabla h(1, -2, 1) = (10, 21, 9),$$

and hence the tangent plane to the surface $\mathcal{S}$ at the point $(1, -2, 1)$ has cartesian equation

$$(10, 21, 9) \cdot (x - 1, y + 2, z - 1) = 0 \quad \Leftrightarrow \quad 10x + 21y + 9z = -23.$$

Exercise 0.5 Consider the function

$$f(x, y, z) = zx^2 + \ln(y^2 + 2z^2 + 1) - e^{xy} - \ln 3.$$

- (a) Write down the cartesian equation of the level surface $\mathcal{S}$ of the function f passing through the point $(1, 0, 1)$.

Hint: $\mathcal{S}$ has cartesian equation: $zx^2 + \ln(y^2 + 2z^2 + 1) - e^{xy} - \ln 3 = f(1, 0, 1)$.

- (b) Write down the cartesian equation of the tangent plane to the surface $\mathcal{S}$ at the point $(1, 0, 1)$.

Hint: the normal vector of the tangent plane is $\nabla f(1, 0, 1)$.

- (c) Write down the parametric equations of the normal line to the surface $\mathcal{S}$ at the point $(1, 0, 1)$ i.e., the line passing through the point $(1, 0, 1)$ and which is perpendicular to the surface $\mathcal{S}$.

Hint: The normal line has direction $\nabla f(1, 0, 1)$.

**The next set of notes will be on
Maximum-Minimum I: Unconstrained Optimization!**